Case 24: Combined fuel, slope, and wind interactions

1 Verification purpose

This case exercises simultaneous spatial fuel variation, opposing slopes, and steady wind to reveal integration failures that isolated cases may miss. The decision checks the independently derived domain-maximum head spread while preserving the Guide configuration and central ignition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-specific outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

At each active front cell ELMFIRE combines fuel-dependent reaction terms, wind and slope factors, and elliptical propagation geometry to obtain a local normal velocity. The configuration intentionally produces competing directional effects. Although the discontinuous coupled perimeter has no closed-form solution, its maximum local direction-of-maximum-spread response does.

For each active cell, ELMFIRE evaluates fuel-dependent R_0, wind factor ϕ_w, and slope factor ϕ_s. Wind and slope are combined as vectors before the direction of maximum spread and effective-wind ellipse are formed:

$$\boldsymbol{\phi} = \phi_w \widehat{\boldsymbol{u}}_w + \phi_s \widehat{\boldsymbol{u}}_s, \qquad R_H = R_0(1 + |\boldsymbol{\phi}|). \tag{1}$$

Spatial discontinuities in fuel and slope make R_H piecewise and the resulting level-set problem has no closed-form perimeter.

3 Derivation of expected behavior

The dominant generated condition is the southwest quadrant: fuel model 4, 15-degree slope, 5 mph wind, and fixed 6/7/8 percent dead moisture. The tracked fuel table gives $R_0 = 1.493256$ m/min and wind-adjustment factor 0.547659. The independent Rothermel calculation gives $\phi_s = 2.010270$ and $\phi_w = 7.366964$. Wind-to south and upslope west are orthogonal, so $|\boldsymbol{\phi}| = \sqrt{\phi_s^2 + \phi_w^2}$ and $R_H = 12.896233$ m/min.

The generated Fuel Model 4 and moisture inputs first give the no-forcing rate

$$R_0 = 1.493256 \text{ m min}^{-1}$$

and reaction intensity $I_R = 2272.300$ kW m^{-2}. Independent wind and slope calculations give $\phi_w = 7.366964$ and $\phi_s = 2.010270$. Because the imposed wind and upslope vectors are orthogonal, their dimensionless resultant is

$$\phi_{\text{res}} = \sqrt{\phi_w^2 + \phi_s^2} = \sqrt{(7.366964)^2 + (2.010270)^2} = 7.636318.$$

The reference maximum rate is consequently

$$R_{\text{ref}} = R_0(1 + \phi_{\text{res}}) = 1.493256(1 + 7.636318) = 12.896233 \text{ m min}^{-1}.$$

The decision metric substitutes this value in $e_R = |R_{\text{max}} - 12.896233|/12.896233$ and compares e_R with the tabulated relative tolerance.

4 Verification metrics and acceptance criteria

The required metric is the maximum finite spread rate over the completed domain. It must agree with the independently derived 12.896233 m/min southwest-quadrant head response within the Guide prescribed 10 percent tolerance. The case is NOT EVALUABLE only when required output is missing; otherwise this quantitative metric decides PASS or FAIL.

5 Simulation configuration and predicted outcome

The 30 m domain combines the CASE23 fuel quadrants, CASE22 opposing slopes, 5 mph wind, uniform moisture, and central ignition. Predicted output is asymmetric, spatially continuous, and preferentially extended by both wind and locally faster fuels/slopes.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 shows the prescribed spatial condition used in this experiment. The displayed field represents the initial level-set geometry or a controlling fuel, terrain, moisture, wind, or canopy quantity, as identified by the figure labels. The plotted values come from the prepared simulation inputs, not from the expected response.

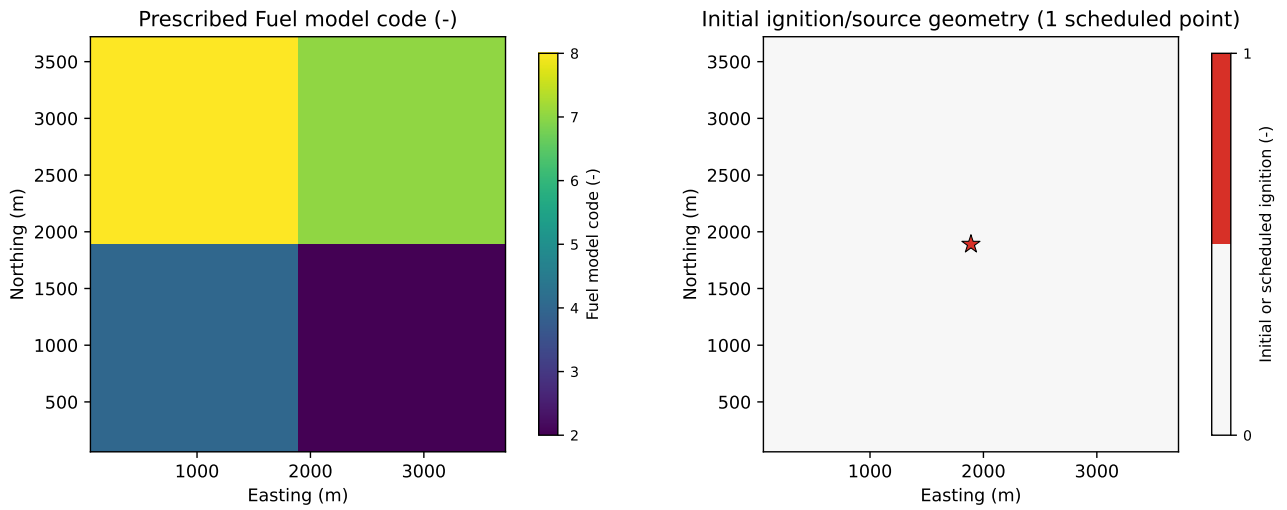


Figure 1: Prepared input conditions from the prescribed spatial fields. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-specific ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 simulation condition(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

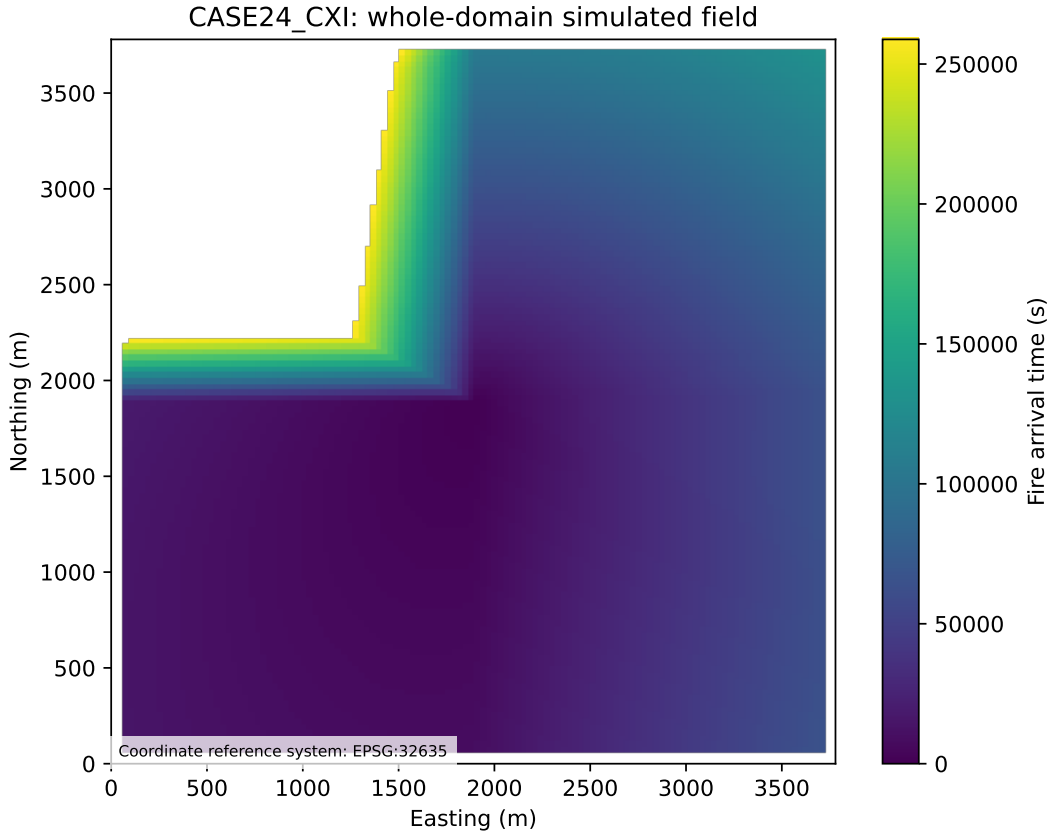


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed simulation condition, in seconds. The coordinate reference system is stated in the figure. The calculated metrics, not visual appearance, determine acceptance.

7 Calculated metrics and verification decision

Simulation condition / metric	Expected	Calculated	Status
main / head ROS	12.8962 m/min	12.8962 m/min	PASS

The displayed decision is **PASS**. A present case-specific calculated metrics controls the table; missing required output remains NOT EVALUABLE.

8 Technical interpretation

Agreement supports the coupled fuel-table selection, wind adjustment, slope factor, vector magnitude, and spread-rate output for the dominant generated condition. A plausible perimeter alone is not used as evidence.

9 Scope and limitations

The metric verifies the analytically observable domain maximum, not the complete discontinuous perimeter or every regional direction. It uses deterministic implementation verification rather than

field validation.